
Physical Hazards in Occupational and Environmental Health

How professionals recognize, evaluate, and control physical energy

Senior-undergraduate lesson • 50–60 minute adaptable baseline

Project-generated for potential guest instruction; not an official Brandeis University lesson and not previously delivered.

The goal is judgment, not a table of limits

Trace energy from source to person

Choose evidence that matches the decision

Separate law, proposal, and recommendation

Prioritize controls and defend a decision

Start with the work, not the hazard label

A maintenance team works in direct sun beside a gasoline-powered compressor. The task requires heavy effort and repeated impact-tool use.

Is the work adequately controlled?

Potential exposures: heat • noise • vibration • radiant/UV energy

What would you want to know first?

WORK

What task and duration?

ENERGY

What source and pathway?

PEOPLE

Who is exposed?

EVIDENCE

What can we measure?

AUTHORITY

What applies?

ACTION

What should change?

Add one question in chat or on the shared board.

Physical hazards transfer energy or impose a condition

ACOUSTIC
Noise

THERMAL
Heat / cold

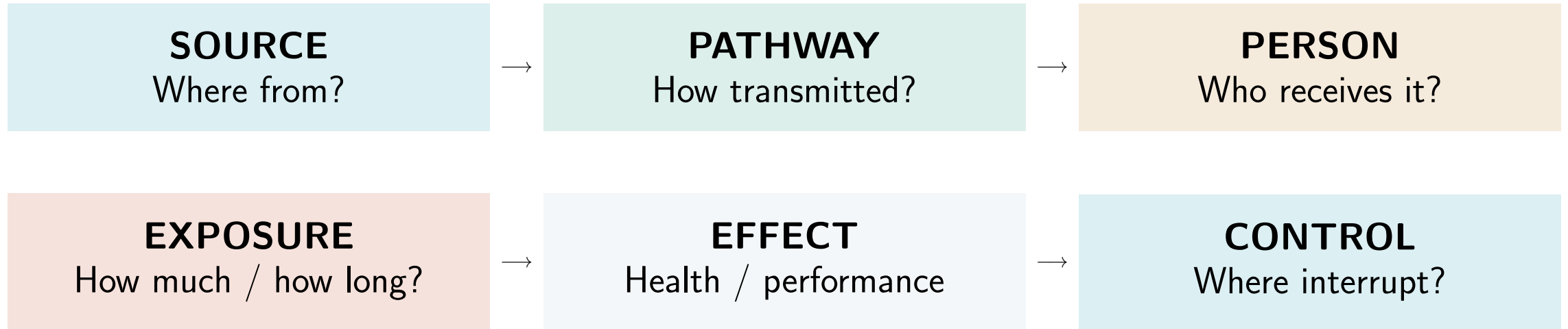
ELECTROMAGNETIC
Radiation

MECHANICAL
Vibration / pressure

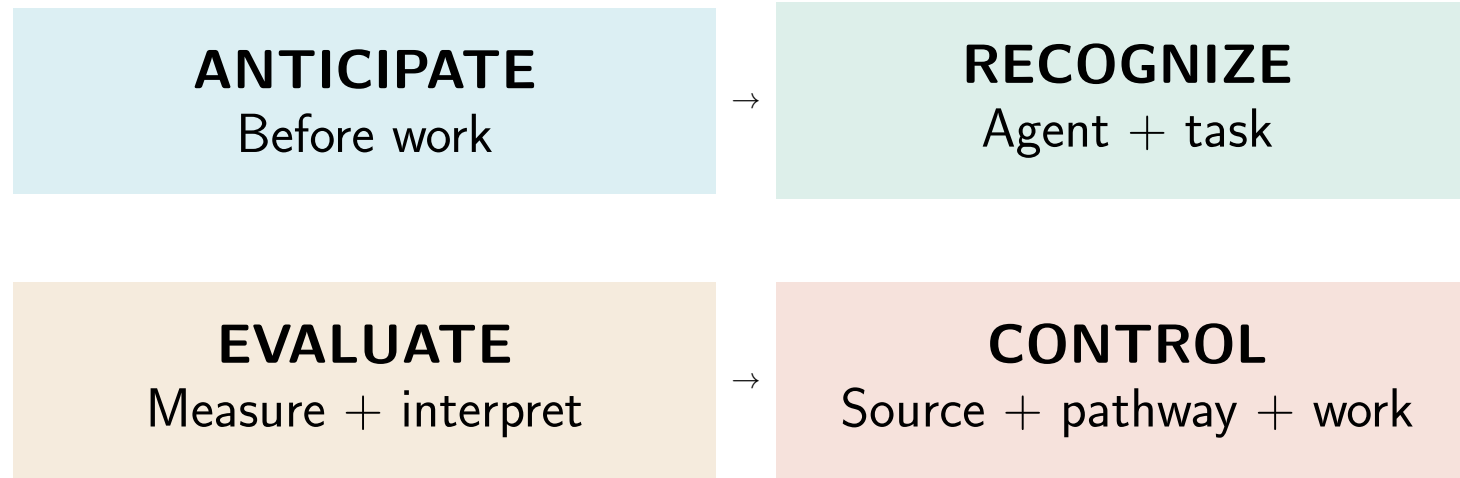
OPTICAL
Light / lasers / UV

The label starts the assessment. It does not finish it.

Follow the energy from source to control



Industrial hygiene turns questions into action



Professional judgment connects the evidence to a bounded recommendation.

Source: OSHA, Industrial Hygiene; NIOSH, Hierarchy of Controls.

A measurement answers only the question it was designed for

A number can show

level, time history, environmental condition, acceleration, or dose

A number may not show

representativeness, worker variability, full-shift exposure, jurisdiction, or control performance

Always record

instrument, placement, settings, calibration, task, time, units, authority, and limitations

Noise case: two readings in the mid-90s dBA

Impact tool at operator position
94–96 dBA

Compressor area
88–91 dBA

Is the worker over a limit?

Missing: duration, mobility, personal dose, settings, calibration, intermittency, and the comparison criterion.
Values shown here are project-generated instructional data, not field observations or limits.

Noise exposure combines level and duration

Sound level + time = exposure evidence

dBA

A-weighted sound level

TWA / dose

Integrated using defined settings

Continuous, intermittent, and impulsive work can require different sampling strategies.

Area and personal measurements answer different questions

SOUND-LEVEL METER

- Screen an area
- Find a source
- Compare locations

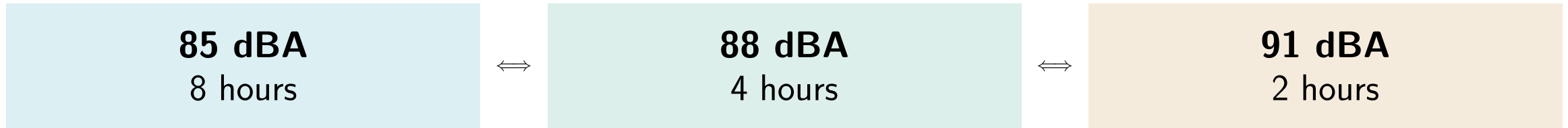
PERSONAL DOSIMETER

- Follow the worker
- Integrate changing levels
- Estimate time-based dose

A point reading is not an eight-hour TWA.

Source: NIOSH, Understand Noise Exposure; OSHA, 29 CFR 1910.95.

NIOSH equal-energy example: time halves every 3 dB



This is a NIOSH health-based recommendation using a 3-dB exchange rate. It is not the OSHA compliance calculation.

Source: NIOSH, Understand Noise Exposure.

The same number can belong to different systems

	Federal OSHA general industry	NIOSH recommendation
Eight-hour basis	90 dBA permissible exposure basis	85 dBA REL
Exchange rate	5 dB	3 dB
85 dBA	Hearing-conservation action level	Recommended exposure limit
Authority	Enforceable when applicable	Health-based recommendation

Never say only “the 85-dBA limit.” Name the issuer, metric, and purpose.

Source: OSHA, 29 CFR 1910.95; NIOSH, Understand Noise Exposure.

Control noise at the source and along the pathway

Quieter process or tool

Enclose, isolate, damp

Maintain and balance

Barrier or distance

Reduce duration safely

Fit hearing protection

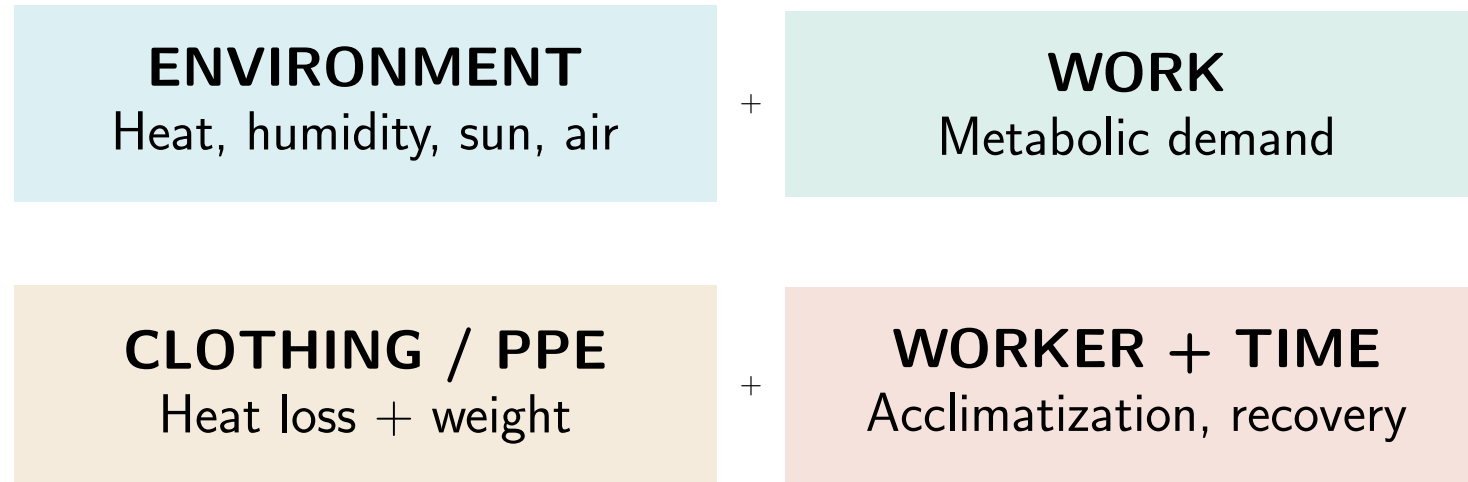
Hearing protection matters—but it is not permission to skip feasible engineering controls.

Heat case: a weather value is not an exposure assessment

Direct sun • heavy work • long sleeves and PPE • new and returning workers • recovery area at a distance

What produces the heat load?

Heat stress is an interaction



Heat stress is the load. Heat strain is the body's response.

Source: NIOSH, Occupational Heat Stress; OSHA, Heat Overview.

WBGT improves environmental assessment—but is not the whole answer

WEATHER / HEAT INDEX

Useful awareness
Not workload or clothing

ON-SITE WBGT

Air + humidity + radiant heat + movement

Interpret with workload, clothing/PPE, acclimatization, time, recovery, and current guidance.

Source: OSHA, Heat Overview.

Acclimatization changes—but does not eliminate—risk

NEW WORKER

Gradual exposure over 7–14 days

RETURNING WORKER

May need reacclimatization

NIOSH's example schedule is a recommendation, not a universal federal legal trigger. Protect workers without demanding unnecessary medical disclosure.

Source: NIOSH, Workplace Recommendations and Acclimatization.

Four labels keep the federal heat landscape accurate

CURRENT AUTHORITY

General Duty Clause and other applicable standards for recognized serious heat hazards

ENFORCEMENT PROGRAM

OSHA CPL 03-00-024, dated April 10, 2026; not a heat PEL

PROPOSED RULE

Federal heat-specific standard remains proposed; no final rule asserted

RECOMMENDATION

NIOSH technical prevention criteria; not OSHA law

Source: OSHA, Heat Standards, Heat Rulemaking, and 2026 Heat NEP; NIOSH, Heat Recommendations.

Heat controls must work as a system

Reduce source or radiant heat

Ventilate, cool, mechanize

Move recovery closer

Adjust workload and schedule

Acclimatize, hydrate, train

Plan emergency response

Define who can stop or modify work—and what triggers that decision.

Radiation is not one homogeneous hazard

IONIZING

X-rays, gamma, alpha, beta, neutrons
Can remove electrons

NON-IONIZING

UV, visible, infrared, RF/microwave, lasers
Source-specific effects

Identify the source, energy, pathway, geometry, detector, task, and regulator before selecting a limit.

Source: OSHA, Ionizing Radiation and Non-Ionizing Radiation.

For ionizing radiation: time, distance, shielding—and ALARA

TIME

Minimize near source

DISTANCE

Increase appropriately

SHIELDING

Match source and energy

ALARA means keeping dose as far below applicable limits as reasonably achievable. It supplements limits; it does not replace them.

Authority depends on the source, workplace, and activity; verify the applicable radiation program.

Source: OSHA, Ionizing Radiation—Control and Prevention.

Vibration enters through the hand—or the whole body

HAND–ARM

Powered tools / workpieces
Vascular + neural effects

WHOLE BODY

Seats / floors / vehicles
Shocks + posture + low-back risk

Acceleration needs axes, frequency weighting, contact point, duration, task, and equipment condition.
Manufacturer emission is not worker exposure.

Source: NIOSH, Hand-Arm Vibration Criteria and 2025 Whole-Body Vibration HHE.

Integrated case: the evidence you have

Work	09:00–12:00; 150 minutes of work; six 10-minute impact-tool cycles assigned to one worker
Noise	Operator-position screening: 94 and 96 dBA; compressor area: 88–91 dBA; no personal dosimetry
Heat	WBGT 29.5 degrees C at 09:20 and 31.0 degrees C at 10:35; direct sun; light air movement
Demand	Heavy during tool/fastener work; moderate during setup and inspection
Recovery	10 minutes after each 50 minutes; shade about 80 meters away

All names and values are project-generated instructional data. They are not observations or exposure limits.

Integrated case: the uncertainty you must manage

People	Two routinely work in heat; one returned after 10 days away; one is new to the task and climate
PPE	Work clothing, gloves, hard hat, eye protection, hearing protectors; fit and heat burden not evaluated
Emergency	Water at vehicle; radio available; team has not reviewed the written heat-emergency plan today
Vibration	Manufacturer lists 4.8 m/s^2 ; field acceleration, axes, grip, maintenance, and duration are unknown
Flexibility	A 30–45 minute delay and task reassignment are available; equipment-relocation feasibility is unconfirmed

The data support precaution and planning—not a complete compliance determination.

Make a decision—and show your reasoning

RECOGNIZE

Priority hazards

EVALUATE

Measurement + uncertainty

CONTROL

Immediate + durable

Three sentences: The evidence shows... The evidence does not establish... We recommend...

Include one verification step and one stop-or-modify-work trigger. More than one answer can be defensible.

Professional judgment is evidence made actionable

- 01 Follow energy from source to person.
- 02 Match measurement to task, time, and decision.
- 03 Name the authority: law, enforcement, proposal, or recommendation.
- 04 Control the source and pathway before relying on PPE.
- 05 State uncertainty, verify performance, and define a trigger.

What would change your recommendation?

Appendix: official sources and point-of-use currentness

Noise: OSHA, [29 CFR 1910.95](#); NIOSH, [Understand Noise Exposure](#).

Heat: OSHA, [Heat Overview](#), [Heat Standards](#), [Heat Rulemaking](#), and [2026 Heat NEP](#); NIOSH, [Workplace Recommendations](#).

Radiation: OSHA, [Ionizing Radiation—Control and Prevention](#) and [Non-Ionizing Radiation](#).

Vibration: NIOSH, [Hand-Arm Vibration Criteria](#) and [2025 Whole-Body Vibration HHE](#).

Controls: NIOSH, [Hierarchy of Controls](#).

Baseline checked 10 August 2026. Recheck the heat rulemaking, jurisdiction, state-plan requirements, host context, and numerical criteria before delivery. The case is synthetic.